

STRASSEN'S MATRIX MULTIPLICATION

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METHOD 1: GENERAL MATRIX MULTIPLICATION

Given two square matrices A & B of size $n \times n$ each,

find their multiplication matrix.

* It is based on the row-by-column rule.

* Here A & B are two matrix & result stored in ' C ' matrix

$$C = AB$$

* Formula: Each element of C is calculated as:

$$C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$\therefore$ row i of A is multiplied with column j of B

Eg: 2×2 Matrix multiplication

Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$C = AB$$

Here these 2×2 matrix is divided into size $n \times n$

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} (a_{11} * b_{11}) + (a_{12} * b_{21}) & (a_{11} * b_{12}) + (a_{12} * b_{22}) \\ (a_{21} * b_{11}) + (a_{22} * b_{21}) & (a_{21} * b_{12}) + (a_{22} * b_{22}) \end{bmatrix}$$

$\therefore$ These 4 formula are derived to do 2×2 multiplication

$$C_{11} = (a_{11} * b_{11}) + (a_{12} * b_{21})$$

$$C_{12} = (a_{11} * b_{12}) + (a_{12} * b_{22})$$

$$C_{21} = (a_{21} * b_{11}) + (a_{22} * b_{21})$$

$$C_{22} = (a_{21} * b_{12}) + (a_{22} * b_{22})$$

∴ General multiplication requires:

→ 8 scalar multiplication

→ 4 scalar additions.

Calculate nxn matrices time complexity based on Algorithm

Algorithm matrix (A, B, C, n)

{

for (i=0; i<n; i++)

→ (n+1)

{

for (j=0; j<n; j++)

→ n(n+1)

{

C[i, j] = 0

→ n * n

for (k=0; k<n; k++)

→ n * n * (n+1)

{

C[i, j] += A[i, k] * B[k, j] → n * n * n

}

}

}

$$\begin{aligned} &= (n+1) + n^2 + n + n^2 + n^3 + n^2 \\ &\quad + n^3 \\ &= 3n^2 + 2n^3 + 2n + 1 \\ &= 2n^3 + 3n^2 + 2n + 1 \end{aligned}$$

$$T(n) = O(n^3)$$

Time complexity for two $n \times n$ matrices

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$$T(n) = O(n^3)$$

Method 2: Divide and conquer Matrix multiplication

Divide and conquer is a method of multiplying matrices by dividing each matrix into smaller submatrices, recursively multiplying them, and then combining the results.

* Following 4×4 Matrix multiplication using Divide and conquer.

eg: Suppose we have 4×4 matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$n \times n \Rightarrow 4 \times 4$

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{bmatrix}$$

$n \times n \Rightarrow 4 \times 4$

We want to calculate:

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix}$$

$$C = A \times B$$

Step 1: Divide the 4×4 matrices

Divide each matrix A & B into '4' (2×2) submatrices of size $n/2$ & $n/2$.

$$\begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{bmatrix}$$

$\therefore a_{11}, a_{12}, a_{21}, a_{22}$ are of sub matrix of a size $n/2 \times n/2$

$$A \rightarrow n/2 \times n/2 \rightarrow 4/2 \times 4/2 \Rightarrow 2 \times 2 \text{ matrix}$$

$\therefore 4 \times 4$ matrix is reduced to 2×2 matrix.

$\therefore$ Big problem is reduced to smaller problem.

where:

* A_{11} is assumed as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

* $\begin{bmatrix} a_{13} & a_{14} \\ a_{23} & a_{24} \end{bmatrix}$ assumed as A_{12}

$$A_{21} = \begin{bmatrix} a_{31} & a_{32} \\ a_{41} & a_{42} \end{bmatrix} \quad A_{22} = \begin{bmatrix} a_{33} & a_{34} \\ a_{43} & a_{44} \end{bmatrix}$$

Similarly $B_{11} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, B_{12} = \begin{bmatrix} b_{13} & b_{14} \\ b_{23} & b_{24} \end{bmatrix}$

$$B_{21} = \begin{bmatrix} b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix} \quad B_{22} = \begin{bmatrix} b_{33} & b_{34} \\ b_{43} & b_{44} \end{bmatrix}$$

$$\text{Similarly } C_{11} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}, \quad C_{12} = \begin{bmatrix} c_{13} & c_{14} \\ c_{23} & c_{24} \end{bmatrix} \quad (3)$$

$$C_{21} = \begin{bmatrix} c_{31} & c_{32} \\ c_{41} & c_{42} \end{bmatrix} \quad C_{22} = \begin{bmatrix} c_{33} & c_{34} \\ c_{43} & c_{44} \end{bmatrix}$$

$$\therefore C = A B$$

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Therefore 2×2 matrices formula is applied here.

Step 2:- conquer :- calculate the four 2×2 blocks of C :

$$C_{11} = A_{11}B_{11} + A_{12}B_{21}$$

$$C_{12} = A_{11}B_{12} + A_{12}B_{22}$$

$$C_{21} = A_{21}B_{11} + A_{22}B_{21}$$

$$C_{22} = A_{21}B_{12} + A_{22}B_{22}$$

} Combine the four 2×2 result matrices and stored in 'C' matrix.

$\therefore$ The Recurrence Relation for Matrix multiplication

using DAC is

$$T(n) = \begin{cases} 1 & n < 2 \text{ // small problem} \\ 8T(n/2) + n^2 & n > 2 \end{cases}$$

At each level:

→ There are 8 recursive multiplications

→ Matrix additions require $O(n^2)$ time

Calculate Time complexity

$$T(n) = 8T(n/2) + n^2$$

$$a = 8 \cdot b = 2 \quad f(n) = n^2$$

Using the Master Theorem:

Case (iii): $a > b^d$

$$8 > 2^1,$$

$$8 > 2 [T]$$

Apply Case '3' time complexity, then

$$T(n) = O(n^{\log_b a})$$

$$= O(n^{\log_2 8})$$

$$= O(n^{\log_2 2^3})$$

$$= O(n^{3 \log_2 2})$$

$$T(n) = O(n^3)$$

$\therefore$ Time complexity for matrix multiplication using DAC is $O(n^3)$

Note: The no. of multiplications is still 8. so its time complexity remains $O(n^3)$.

$\Rightarrow$ It does not improve the asymptotic complexity over general matrix multiplication

METHOD 3: STRASSEN'S MATRIX MULTIPLICATION

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DEFINITION:

- * Strassen's matrix multiplication using divide and conquer technique.
- * That divides matrices into smaller submatrices and reduces the number of recursive multiplications from 8 to 7.
- * Hence overall time complexity reduced.

∴ Strassen's time complexity for a matrix size $n \times n$ is:

$$O(n^{\log_2 7}) = O(n^{2.807})$$

Step 1: Divide the matrices:

for 2×2 matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

We want to find:

$$C = AB$$

$$\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} (a_{11} * b_{11}) + (a_{12} * b_{21}) & (a_{11} * b_{12}) + (a_{12} * b_{22}) \\ (a_{21} * b_{11}) + (a_{22} * b_{21}) & (a_{21} * b_{12}) + (a_{22} * b_{22}) \end{bmatrix}$$

In general method it requires 8 multiplications, 4 additions.

- * In Strassen's reduces this to 7 multiplications.

Step 2: calculate seven products;

Strassen's defines the following seven products.

* By using formula, we can reduce the time complexity for matrix multiplication using Strassen's DAC.

$$P = (A_{11} + A_{22})(B_{11} + B_{22})$$

$$Q = B_{11}(A_{21} + A_{22})$$

$$R = A_{11}(B_{12} - B_{22})$$

$$S = A_{22}(B_{21} - B_{11})$$

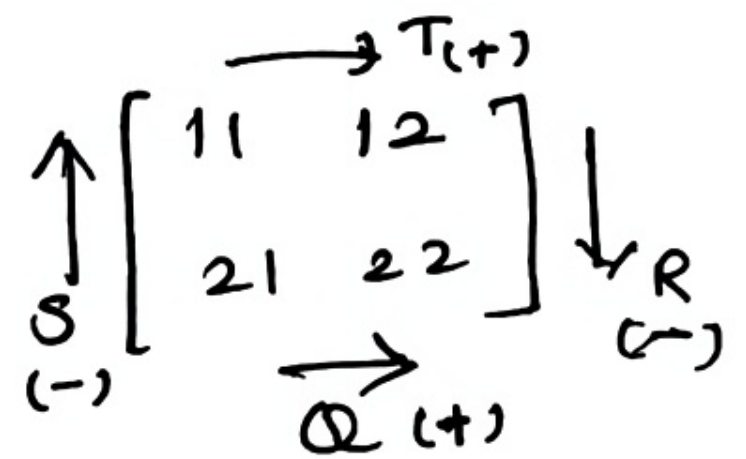
$$T = B_{22}(A_{11} + A_{12})$$

$$U = (A_{21} - A_{11})(B_{11} + B_{12})$$

$$V = (B_{21} + B_{22})(A_{12} - A_{22})$$

NOTE: TRICK

$$B_{11} \quad A_{11} \quad A_{22} \quad B_{22}$$



∴ It requires

$$\text{Multiplications} = 7$$

Step 3: Combine the results

The four elements of the resultant matrix 'C' are calculated as:

$$C_{11} = P + S - T + V$$

$$C_{12} = R + T$$

$$C_{21} = Q + S$$

$$C_{22} = P + R - Q + U$$

NOTE:

RAT

∴ Hence

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

$$C = \begin{bmatrix} (P+S-T+V) & (R+T) \\ (Q+S) & (P+R-Q+U) \end{bmatrix}$$

* It performs multiplications = 7

* Total Additions = 18

RECURRENCE RELATION FOR STRASSEN'S MATRIX MULTIPLICATION

$$T(n) = 7T(n/2) + O(n^2)$$

$$T(n) = 7T(n/2) + 18n^2 \quad n \geq 2$$

$O(n^2)$ = total time taken to compute Addition & Multiplication

$$\therefore T(n) = 7T(n/2) + n^2 \quad n \geq 2$$

Calculate Time complexity:

Using Master theorem:

$$a = 7 \quad b = 2 \quad f(n) = n^2$$

$$\text{Case (iii)} \quad a > b^d \quad d = 1$$

$$7 > 2^1$$

$7 > 2 [7] \therefore$ Apply case '3' Time complexity

$$T(n) = O(n^{\log_b a})$$

$$= O(n^{\log_2 7})$$

$$T(n) = O(n^{2.81})$$

$\therefore$ Time complexity reduced to $O(n^3)$ to $O(n^{2.81})$

$\therefore$ Strassen's matrix multiplication time complexity is

$$T(n) = O(n^{2.81})$$

Time complexity for Strassen's:

1. Best case: $O(n^{2.81})$

2. Average case: $O(n^{2.81})$

3. Worst case: $O(n^{2.81})$

Eg: If matrices $A = \begin{bmatrix} 9 & 4 & 6 & 7 \\ 7 & 8 & 1 & 4 \\ 4 & 3 & 2 & 6 \\ 5 & 3 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 7 & 6 & 2 & 1 \\ 3 & 9 & 0 & 3 \\ 2 & 5 & 2 & 9 \\ 3 & 2 & 4 & 7 \end{bmatrix}$

Implement Strassen's matrix multiplication on A & B.

Sol:

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